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Deep Learning Based Hand Gesture Translation System

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Abstract—

Sign language plays an important role for the people who have the hearing and speech problems For the Non- verbal communication between the deaf-mute people. For the Kinetics gestures plays a crucial role in our day-to-day life. There are different sign languages that are available Hand Gesture Translation is one among them. The sign language is a collection of different Hand Gesture symbols and Each Hand Gesture symbol have special meaning. It is mostly used by the deaf-mute persons to express their views and thoughts very quickly with the normal people. Main problem with this system is very difficult to translate the symbols and required special training on sign language. To overcome this problem we have implemented a Hand Gesture Translation System it provides an ability to interact with the machine efficiently. It will help the deaf-mute to express their feelings and views more effectively with the normal people. In this paper implemented software prototype that will automatically recognize the hand gestures with an accuracy of 93.4% for gesture Translation which will help the deaf and dumb people to interact easily with the normal people. The main aim of the project is to provide the easy way of communication between the normal people and the deaf-mute through gestures.

Keywords—Hand Gestures; Deep learning; Machine learning; Pre-Processing Image; Feature Extraction of Image; Image Processing; Deaf-mute.

I INTRODUCTION

The Sign language plays an vital role for the communication between the deaf-mute. Hand Gesture Translation System(HGT System) is one among them, which will help the deaf and dumb people to communicate with the normal people through different hand gestures. These hand gestures will help the deaf and dumb people to express their views and thoughts very quickly with the people. The output is produced in the form text so that the people can recognize what the deaf and dumb are saying. The sign language mostly reduces the communication gap between the deaf-mute and the other people.

There are two types of sign languages that are available one is Image based sign language and another one is Sensor based sign language. Sensor based sign language is very costly compared with the Image based sign language because Sensor based sign language is built with the Hardware components whereas Image based sign language is built through Camera, Our Laptop consists of inbuilt Camera which will help to capture the image objects very easily. Hand Gesture Translation System is the software prototype it uses the Laptop inbuilt camera for capturing the different hand gestures and produces the output in the form of Text. It is mostly used by people who have speech and hearing problems to speak with the people. Because sometimes the people cannot understand what deaf and dumb people are saying in that case these hand gesture Translation system will used because the output is produced in the form of text, which the normal people can able to read. One of the Advantage of Image based approach is not to wear the Hand Gloves, Helmet etc. Sign language identification is significant in many domain areas such as user-interface, Interaction, security and multimedia. There are two parts in sign language identification one is sign detection and another one is sign Translation. The system will use the web camera for capturing the hand gestures and after successfully capturing it will recognize those hand gestures and produces the output in the form of text.

In Machine Learning is useful in solving different real-time problems. This technology is mostly performs complex jobs such as classifying data, Translation of data, identifying data and predict the values. The basic idea of every machine learning project is to give a input data to the machine to generate a system which can be produce the result. Thus the obtained result offer the correct response with the new input or generate predictions for the known information. The objective of our proposed system is to train ML algorithm in order to classify the different hand gesture images such as palm, fist. This method is also using deep learning and Tensor flow frame work .

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Word HGT System is done by deep learning using neural networks. [6] It uses a classification technique which comes under supervised learning. Supervised learning is the function which is trained with the named dataset. The named dataset contains an input-output pairs. In this method the training as well as test datasets required. Classification algorithms are used for prediction in machine learning and works with labeled datasets. In classification, the computer program must train with the training dataset and after the training dataset will be divided into different classes.

Deep learning is a one of machine learning algorithm utilizes multiple layers that process the data and also extract the high level features from them and produces a mathematical model[7]. The proposed system is able to categories the different hand signs, i.e A system, can learn the meaning of each sign and classifies them properly.

Designing a model:

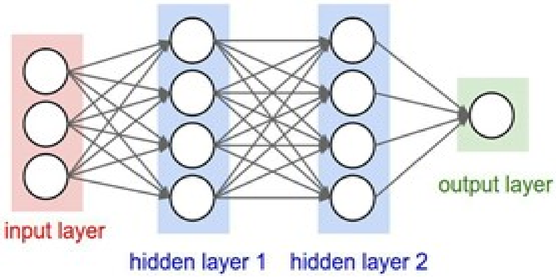


Fig:1.1 Designing a model

Convolutional layer will uses various of filters to the picture, the layer plays out certain numerical tasks in order to deliver one result. [8]Pooling layers are utilized in CNN to reduce the dimensionality of the picture. In a thick layer each hub in the layer is associated with the other hub in the first layer.

II Literature survey

Intrinsically there are two ways for the signal language understanding, one is vision based Translation and another one is sensor based gesture Translation. Sensor based approach includes wearing a helmet , wires etc. and lots of research should be done for the sensor based approach and are very costly. Also there is a disadvantage in the sensor based approaches is that continuously wearing of helmet and wires are impossible, so proposed project is focused on image processing methods Maintaining the Integrity of the Specifications[1]

The In last few years there is a lot of work is done on gesture Translation and sign language Translation [2]. There are many paths or ways for sign Translation which includes Hidden marcov model, Artificial neural network, Eigenvalue based Algorithm are one of the conspicuous accomplishments that were accomplished in the field of Hand gesture Translation is to remove correspondence hindrance between the deaf-mute and the ordinary individuals[3][4][5].

After searching for different datasets in the internet, we did not find the dataset. To search of the dataset is also one of the biggest task, once we search for the dataset it is very easy to build the model according to our dataset. The sign language contains huge signs, hence we need a so much for conversion. Here we build a simple model using conventional neural network concept in deep learning. The architecture is to order the static pictures, we got a decent precision, after that we chose to add more experiments to our model, we additionally confronted a few requirements of the acknowledgment of our undertaking.

III Feasibility Study

The aftereffects of practicality investigation of a deep learning scheme for gesture based communication movement Translation are given in this paper. For motion Translation Deep learning and conventional classification schemes were utilized, and there are looked at[14][15]. In a Deep learning measure each edge of hand motion will be passed for include extraction. After extracting the feature it will recognize the extracted feature and produce the output[12][13].

At this stage the design of the neural network and discretionary boundaries for deep learning have not been improved, it is checked that the precision of acknowledgment went from 97.2% for six signals. Though the performance of the conventional schemes are inferior, it is viewed as that these outcomes demonstrate the possibility of utilizing deep learning scheme for hand gesture identification

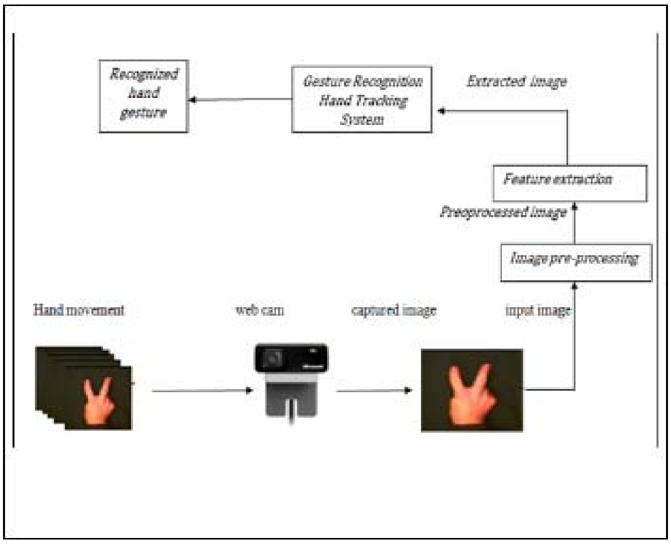
Problem Definition

Sign Language, which will convert the standard Sign Language to Text. This is a multi-use application for deaf and dumb people, which includes many forms of Sign Language and Gestures.

Primarily two methods are used for sign Identification Sight-based and sensor-based Gesture Translation. Sensor-based methodologies like keeping gloves, wires, caps, and so on There is disadvantage in sensor based is wearing a gloves continuously is not possible. So image based is the simple technique for capturing the hand images[9][10][11].

VI Proposed System

Fig:4.1: Working Flow of the system



In The proposed system we are having following modules

i) Collect the Image :

deaf-mute person has to sit in front of web camera and start communicating with sign language. Where web camera capture the images and forwarded to application.

ii) Pre-Processing Image

All the images are collected from web camera and these

images are pre-processed noisy data. It will processes

only hand gesture Image

iii) Feature Extraction of Image

The hand Image received from Image Pre-processing features of the Image is extracted such as sign represented by hand and these features are forwarded to Gesture Translation Hand Tracking system.

iv) GRHT system

GRHT system is core of the system and it is a Deep learning application which contains the trained data sets to recognize the hand gesture. To train the data sets we use the Custom Gesture dataset of 6 gesture classes, individually intended for man- machine interfaces(MMI).

GRHT system receives the data for feature extraction module and compares the hand gesture with available data in Input-output data set. If hand gesture matches with any of data in the data set the respective meaning of the hand gesture will be displayed on the screen to other people

V Results

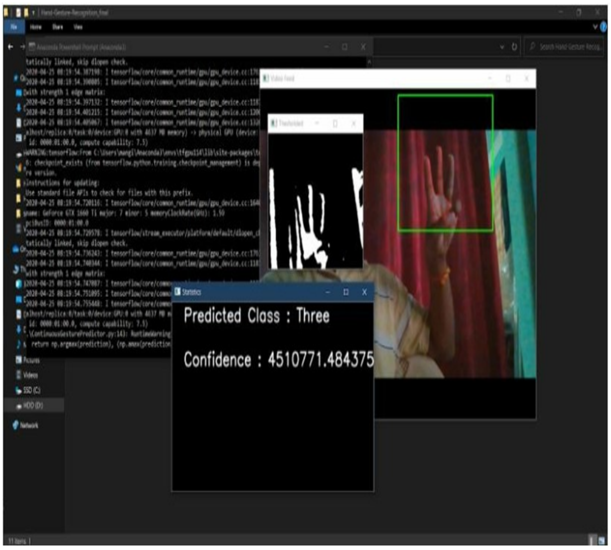


Fig 5.1 Screen short of working of application-1

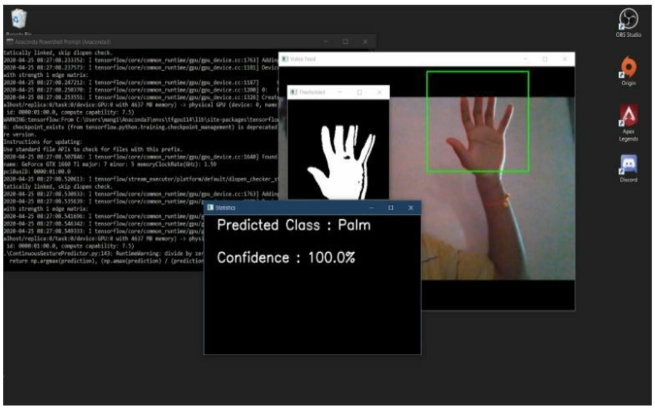


Fig 5.2 Screen short of working of application-2

Use The Custom Gesture dataset of 6 gesture classes, individually intended for man- machine interfaces(MMI) and are recorded by webcam(Internal or external of any frame rate) using background elimination and background masking. There are 7800 images in total, including the testing and the training dataset which are staged at a reasonably dark background. Since the gesture images are recorded by using different webcams, there will be a change in the size of frame, so we cropped them to a standard size. The deep learning model is fed with 6000 images in total, 1000 respectively for each class and is validated with 600 images, 100 respectively for each class.

Although this training set is a lot smaller compared to other datasets that are used in deep learning, using the background elimination concept helped us to remove the over fitting repercussion considerably. The state-of-art model that we used consists of 7 Conv3D layers and a max pooling layer associated with each layer. The architecture is also accommodated with a dropout of 0.75 to avoid over fitting. The final layer consists of the linear activation function, yet very commendable, the soft max is used. The approach we used performs state-of-art performance. The real time video is instantiated using the open cv module and each frame is collected, preprocessed and sent to the model for prediction. Human accuracy is reported as 93.4% for gesture Translation.

Limitations and Future Enhancement

Limitations:

It Replace mouse and Key board , can also Pickup and manipulate Virtual objects Navigate in virtual Environment

Future Enhancement:

It Enhance the Translation capability for various lighting conditions. we can achieve more accuracy by implementing and identifying more number of gestures applying gesture Translation for accessing internet applications

VI Conclusion

This Project is very useful for many people who have the hearing and the speech problems. This project is mainly useful for the deaf and dumb people who are unable to speak and are unable to listen, it will helpful for them.

Sign languages are mostly used by the deaf and dumb people. These sign languages are mostly used to bridge the communication gap between the deaf-mute people and the Individuals. Gestures are used to provide a way for the computers to understand the human body language. Gestures will also play an important role for the communication between the visually impaired people with the normal people.

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